

MZI Design Proposal

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Design and Simulation of an Unbalanced Mach-Zehnder Interferometer on an SOI Platform

Abstract

This project presents the design, simulation, and layout implementation of an unbalanced Mach-Zehnder Interferometer (MZI) using a Silicon-on-Insulator (SOI) photonic platform. The device was designed to operate at a wavelength of 1550 nm with TE polarization and utilizes directional couplers for optical power splitting and recombination. An optical path length difference of approximately 178 μm was introduced between the interferometer arms to generate wavelength-dependent interference. The design was analyzed using Ansys Lumerical MODE and INTERCONNECT, while the layout was developed using KLayout. Simulation results including mode analysis, effective index extraction, transmission spectrum, and free spectral range (FSR) are presented and discussed.

1. Introduction

Silicon photonics has emerged as a promising technology for realizing compact and high-performance optical integrated circuits. Among the various photonic building blocks, the Mach-Zehnder Interferometer (MZI) is one of the most widely used devices due to its simple structure and versatility.

An MZI functions by splitting an incoming optical signal into two separate paths, introducing a phase difference between them, and subsequently recombining the signals to produce constructive or destructive interference. This interference mechanism enables applications such as filtering, sensing, modulation, and optical switching.

The objective of this project is to design and simulate an unbalanced MZI structure on an SOI platform and evaluate its optical characteristics through numerical simulations.

2. Design Objectives

The main objectives of the project are:

- Design an unbalanced MZI on an SOI platform.
- Support single-mode TE propagation at 1550 nm.

- Implement optical splitting and combining using directional couplers.
 - Introduce a controlled path length difference between the interferometer arms.
 - Evaluate device performance through numerical simulations.
 - Generate a fabrication-ready layout using KLayout.
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3. Theoretical Background

3.1 Mach-Zehnder Interferometer Principle

An MZI consists of:

1. Input waveguide
2. Optical splitter
3. Two interferometer arms
4. Optical combiner
5. Output waveguide

The input optical field is divided into two equal portions by the first coupler. As light propagates through the two arms, a phase difference develops due to differences in optical path length. Upon recombination, interference occurs.

The phase difference is expressed as:

$$\Delta\phi = \frac{2\pi n_{eff}\Delta L}{\lambda}$$

where:

- n_{eff} = effective refractive index
 - ΔL = path length difference
 - λ = operating wavelength
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3.2 Free Spectral Range

For an unbalanced MZI, the Free Spectral Range (FSR) is given by:

$$FSR = \frac{\lambda^2}{n_g\Delta L}$$

where:

- n_g = group index
- ΔL = arm length difference

The FSR determines the spacing between adjacent transmission peaks.

3.3 Output Transmission

The normalized transmission of an ideal MZI is:

$$T = \frac{1}{2}(1 + \cos(\Delta\phi))$$

Constructive interference occurs when:

$$\Delta\phi = 2m\pi$$

Destructive interference occurs when:

$$\Delta\phi = (2m + 1)\pi$$

where m is an integer.

4. Device Design

4.1 Platform

Parameter	Value
Platform	SOI
Silicon Thickness	220 nm
Waveguide Width	550 nm
Operating Wavelength	1550 nm
Polarization	TE

The waveguide dimensions were selected to ensure strong optical confinement and support the fundamental TE mode.

4.2 Interferometer Structure

The proposed MZI consists of:

- Input grating coupler
- Input strip waveguide
- First directional coupler
- Two unbalanced interferometer arms
- Second directional coupler
- Output waveguide
- Output grating coupler

A path length difference of approximately:

$$\Delta L \approx 178 \mu m$$

was introduced between the two arms.

This path imbalance enables wavelength-selective interference and generates a periodic transmission spectrum.

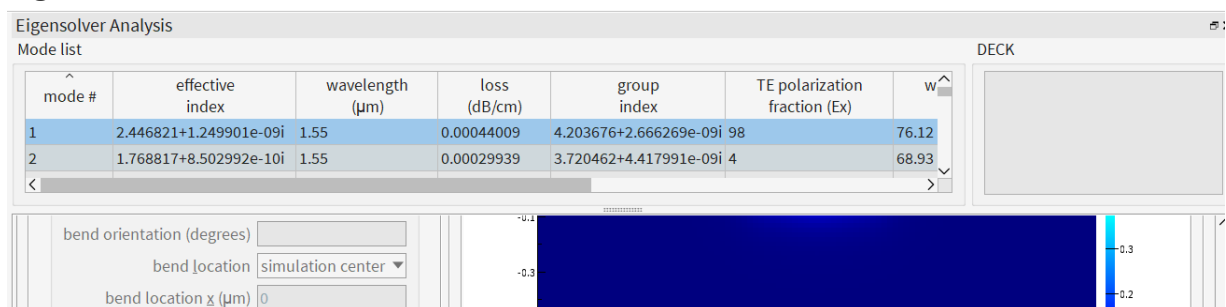
5. Simulation Methodology

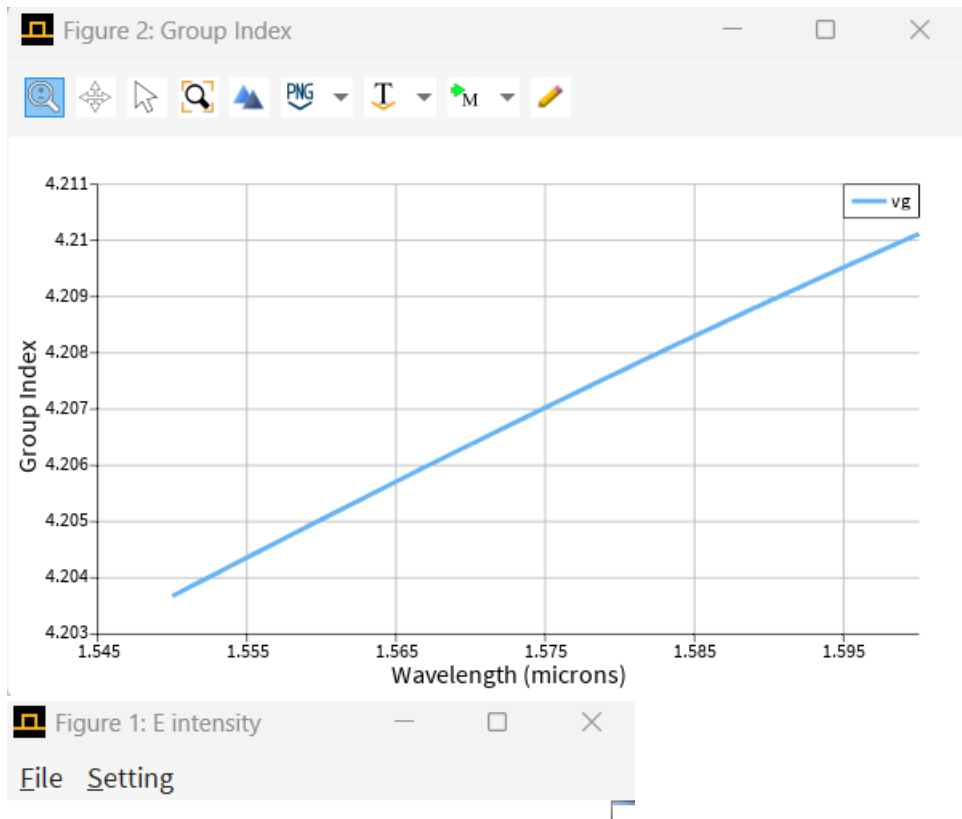
5.1 Mode Analysis

Lumerical MODE was employed to:

- Calculate effective index
- Verify single-mode operation
- Analyze optical confinement
- Extract modal field profiles

Figure :





5.2 Interferometric Analysis

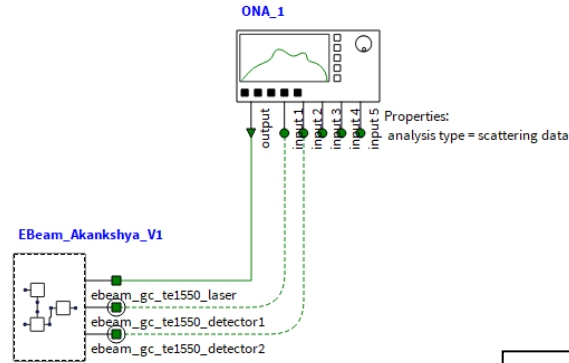
The complete MZI was modeled in Lumerical INTERCONNECT.

The simulation included:

- Directional couplers
- Optical path imbalance
- Waveguide propagation
- Input/output grating couplers

A wavelength sweep was performed around 1550 nm to obtain the spectral response.

Figure :



5.3 FSR Verification

Theoretical FSR calculations were compared with simulation results obtained from the transmission spectrum.

For an unbalanced MZI(For MZI 1),

$$FSR = \frac{\lambda^2}{n_g \Delta L}$$

Parameters :

- $\lambda = 1550\text{nm} = 1.55 \times 10^{-6}\text{m}$
- $\Delta L = 178\mu\text{m} = 178 \times 10^{-6}\text{m}$ (measured in Klayout)

The remaining parameter is the group index n_g .

For a 220 nm × 550 nm SOI waveguide (TE mode), Using $n_g = 4.2$

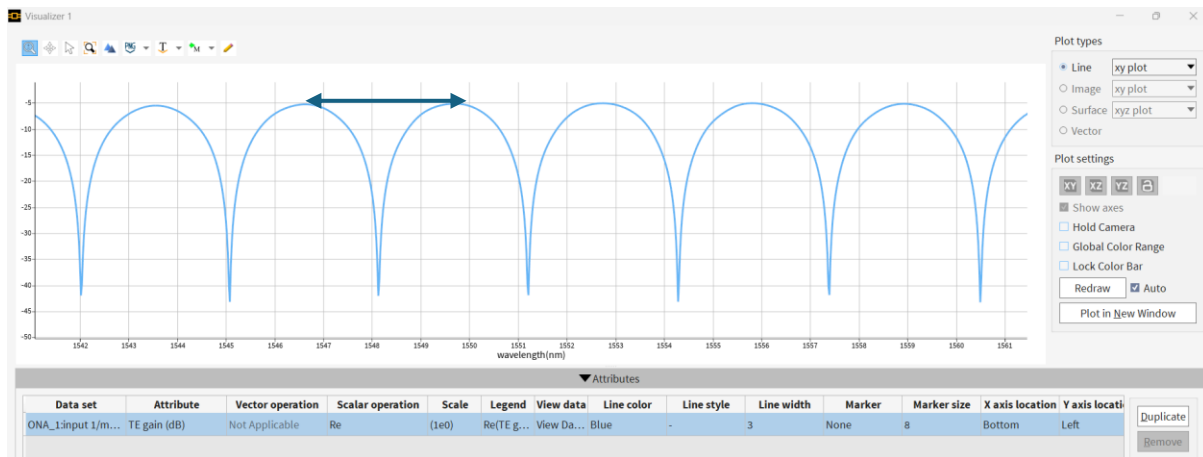
$$FSR = \frac{(1.55 \times 10^{-6})^2}{4.2 \times 178 \times 10^{-6}}$$

$$FSR \approx 3.21 \times 10^{-9} \text{ m}$$

$FSR \approx 3.21 \text{ nm}$

In the simulation also we have got a FSR around 3.1nm.

Figure :



6. Layout Design

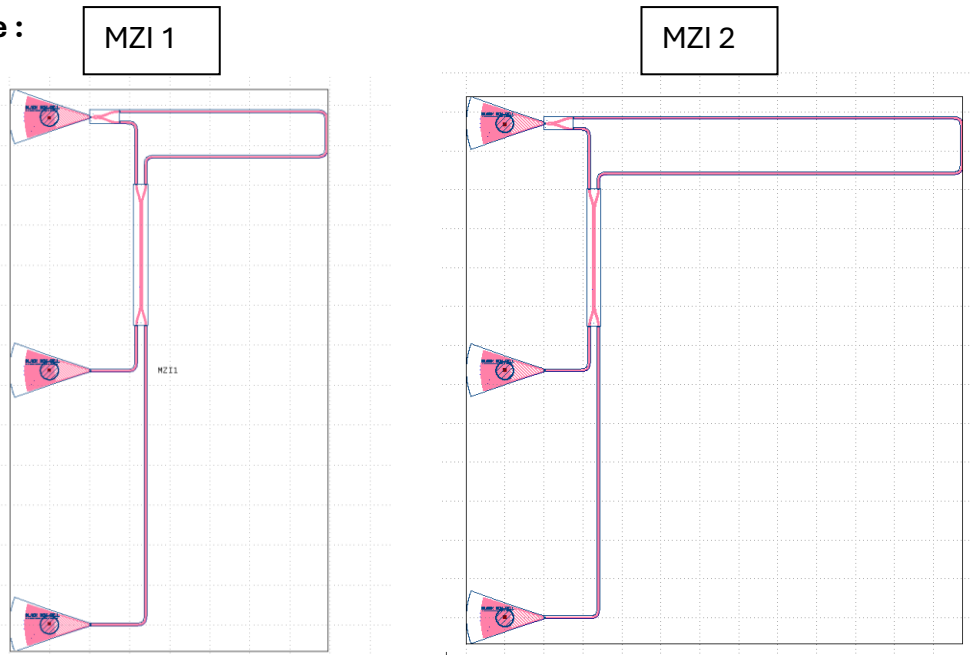
The physical layout of the MZI was created in KLayout.

The design includes:

- Directional couplers
- Routing waveguides
- Input and output ports
- Grating coupler interfaces

Special consideration was given to maintaining smooth bends and minimizing excess optical loss.

Figure :

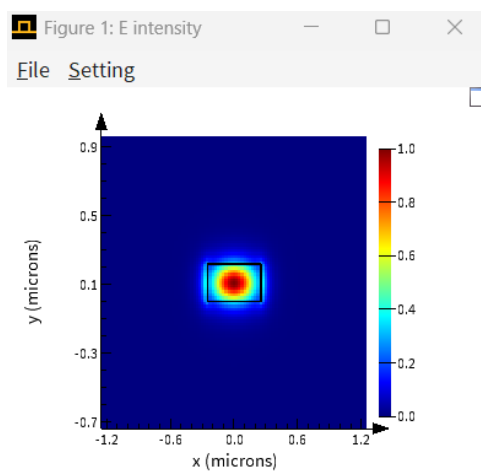
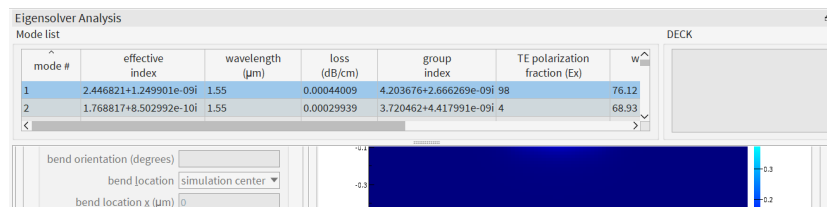


7. Results and Discussion

7.1 Effective Index Analysis

The effective index was 2.44.

Figure:

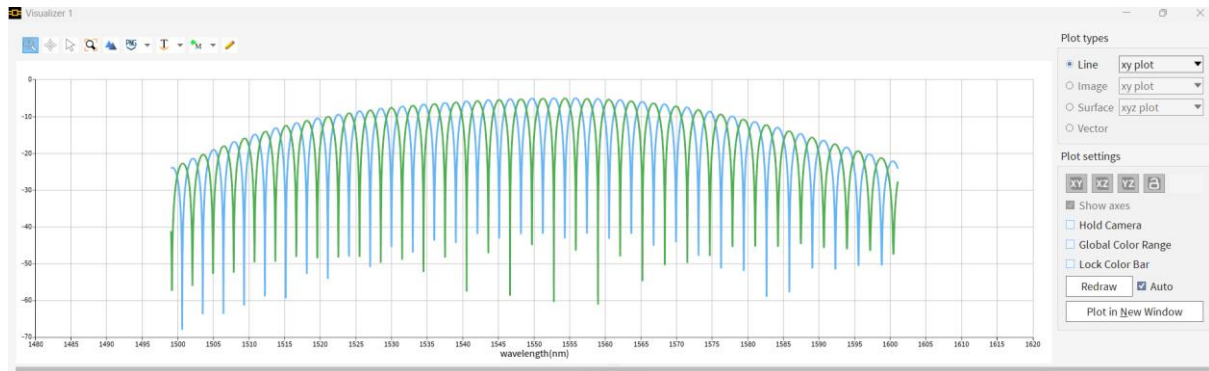


- Optical confinement observed in silicon core.

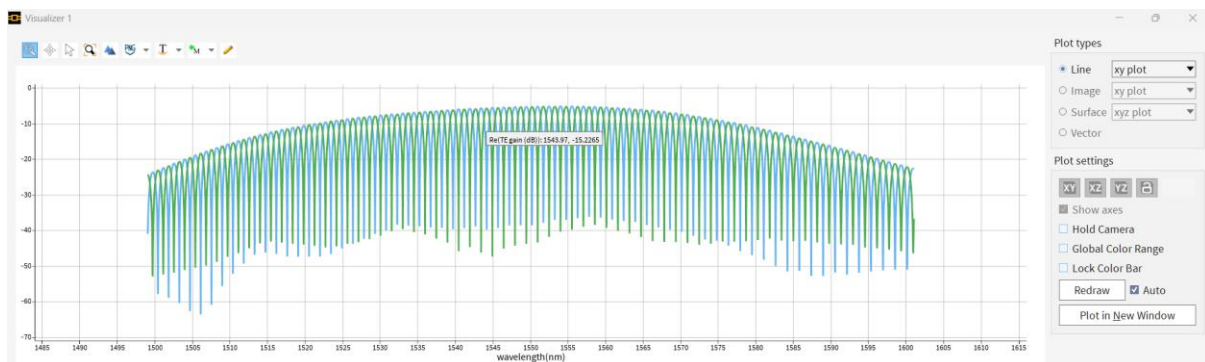
- Fundamental TE mode successfully supported.
- Effective index values validated waveguide operation.

7.2 Transmission Spectrum

Figure:



MZI 1



MZI 2

- Periodic interference fringes observed.
- Constructive and destructive interference confirmed.
- Spectral periodicity matched theoretical expectations.

7.3 Free Spectral Range

Comparison: The FSR Value for the MZI1 is around 3.2nm while in the above figure it can be seen that the FSR Value for MZI2 is different. It is around 2nm. The detailed calculations however I have done for MZI 1.

Parameter	Value
Theoretical FSR	3.2nm
Simulated FSR	3.1nm

7.4 Device Performance

Metric	Value
Effective Index	2.44
Group Index	4.2
FSR	3.1nm

8. Challenges and Design Considerations

Several factors were considered during the design process:

- Ensuring low-loss waveguide routing (Long enough waveguide in klayout to ensure proper bend).
- Selecting an appropriate arm length difference.
- Designing compact layouts in the given floor plan and doing different variants.
- Achieving stable interference characteristics around 1550 nm.

The unbalanced configuration required careful routing to realize the desired optical delay without significantly increasing the device footprint.

9. Conclusion

An unbalanced Mach-Zehnder Interferometer was successfully designed on an SOI platform using 550 nm × 220 nm strip waveguides. The device was designed for TE-polarized operation at a wavelength of 1550 nm and incorporated directional couplers for power splitting and recombination. Numerical simulations were performed using Lumerical MODE and INTERCONNECT, while the layout was developed in KLayout. The obtained results demonstrate the expected interference behavior of the MZI and validate the suitability of the proposed design for integrated photonic circuits.